

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)	Mean of measurements i.e. $x (= 5\Delta y) = 6.6[2]$ [mm] or by impl (1) Mean of $\Delta y = 1.3[2]$ [mm] (1) Absolute uncertainty in $x = 0.3$ [mm] or in $\Delta y = 0.06$ [mm] or by implication (1) % uncertainty = 5 % Accept 4.5 % ecf on Δy and unc in Δy and on x and unc in x (1)		4		4	3	4
		(ii)	Correct substitutions into $\lambda = \frac{a\Delta y}{D}$, with ecf on Δy and allowing slips in powers of 10 (1) $\lambda = 520$ n[m] or 528 n[m] with ecf on Δy (1) % uncertainty = 10 % Accept 9.5 % ecf on uncertainty in Δy (1)		3		3	3	3
	(c)	(i)	$\lambda = \frac{1500 \text{ nm} \sin 45^\circ}{2}$ substitution in re-arranged eq. allow slips of powers of 10 (1) $\lambda = 530$ n[m] Accept 530.3 n[m] (1)		2		2	2	
		(ii)	Bright areas further apart in case of grating (1) Because slits closer together in grating [than double slits] (1) not freestanding Bright areas sharper or smaller or brighter for grating or equivalent (1) Because more slits in grating or more destructive interference or (for brighter fringes) more constructive interference (1) not freestanding	4			4		
			Question 4 total	10	9	0	19	8	7